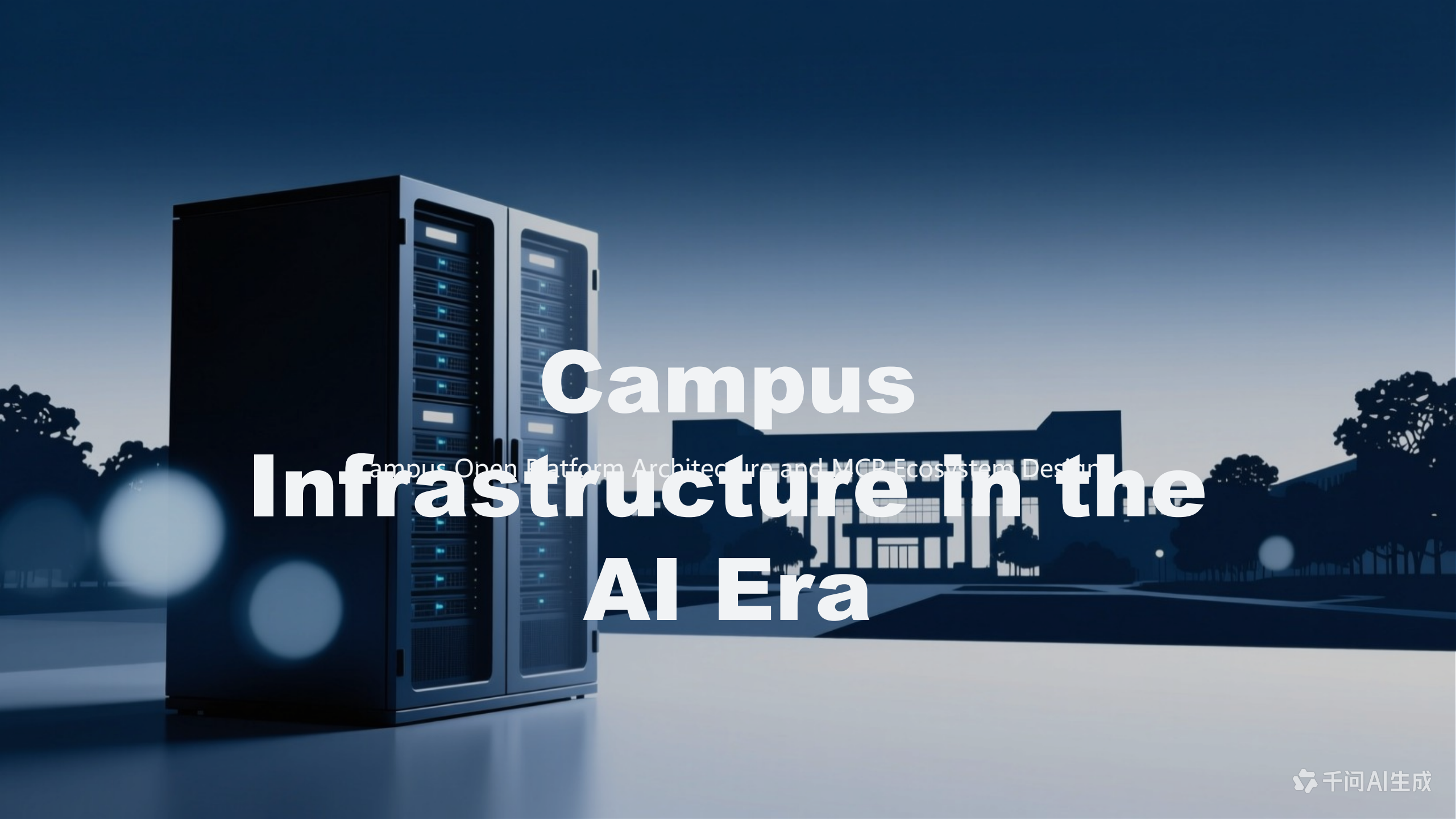


The background of the slide features a large server rack on the left with multiple blue indicator lights. To the right, a modern building with large windows is visible at night, with some trees in the foreground and background. The overall color scheme is dark blue and black.

Campus Infrastructure in the AI Era

Campus Open Platform Architecture and MCP Ecosystem Design

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CHAPTER 01

Background and Challenges of Campus Infrastructure in the AI Era

Paradigm Shift from Information Silos to Intelligent Collaboration

Pain Points of Traditional Campus Systems and

Demands of the AI Era

Traditional campus systems have long suffered from data silos and fragmented interactions, forcing students and faculty to navigate cumbersome cross-system operations. The advent of the AI era demands that campus infrastructure shift from "passive response" to "proactive intelligence," urgently requiring a unified open platform to support



Digital Campus Life Scenarios · Mobile Campus Service Interaction

! Traditional Infrastructure Pain Points

- 01 Systems for academic affairs, student affairs, and finance operate independently with inconsistent data interface standards, causing severe permission barriers and delays in cross-departmental business workflows.
- 02 Students and faculty must frequently switch between multiple graphical interfaces to complete tasks, lacking a unified natural language interaction entry, resulting in extremely high service access costs.

⚙️ AI Era Demands

- 03 Large models require standardized tool invocation protocols to understand and execute campus business rules, demanding that underlying systems expose machine-readable API capabilities.
- 04 Students and faculty expect to accomplish tasks on demand through "single-sentence instructions," requiring the platform to possess intelligent intent recognition and cross-system service orchestration capabilities.

The Inevitable Trend of Leaping from "Digitalization" to "Intelligence"

Smart campus research is undergoing a profound leap from process digitalization to service intelligence. By introducing large models and knowledge graphs, campus infrastructure is no longer merely about data storage and display, but has evolved into an intelligent hub with understanding, reasoning, and execution capabilities, comprehensively

- 01 The maturity of NLP and knowledge graph technologies enables systems to accurately understand complex campus business rules, achieving a shift from "people seeking services" to "services seeking people."

NLP + Knowledge Graph

- 02 The deployment of specialized large models in campus vertical scenarios has significantly improved the efficiency and accuracy of high-frequency scenarios such as official document generation, intelligent form filling, and workload statistics.

Document

- 03 The Ministry of Education's "AI + Education" action plan explicitly proposes building an educational intelligent computing service platform, providing policy guidance for the intelligent upgrading of university infrastructure.

AI + Education



University Library Digital Learning Scenario

Chapter 02

Core Architecture Design of the Campus Open Platform

Building an Integrated, Scalable, Intelligent Hub Foundation

Overall Architecture Blueprint of the Campus Open Platform

The platform adopts a four-layer architecture of "entry - front-end - middle platform - foundation," forming a tightly coupled intelligent system. The middle platform serves as the intelligent hub integrating AI capabilities and MCP toolkits, the foundation provides computing and data support, and the front-end achieves precise service delivery and



Intelligent Entry and Front-end

The unified entry supports PC and mobile terminals, providing policy-based personalized application consoles for precise service delivery.

The front-end flexibly accesses various campus business systems through intelligent interfaces, with intelligent detection and service aggregation mechanisms to effectively bypass faulty nodes.

Unified Entry · Personalized Console



AI Middle Platform and MCP Gateway

The middle platform deploys a large model application gateway, enabling collaboration and on-demand invocation of various AI tools, encapsulating complex business logic.

The introduction of the MCP standard protocol transforms campus system capabilities into toolkits understandable by large models, bridging the interaction link between agents and underlying systems.

Large Model Gateway · MCP Protocol



Computing Foundation and Data Lake

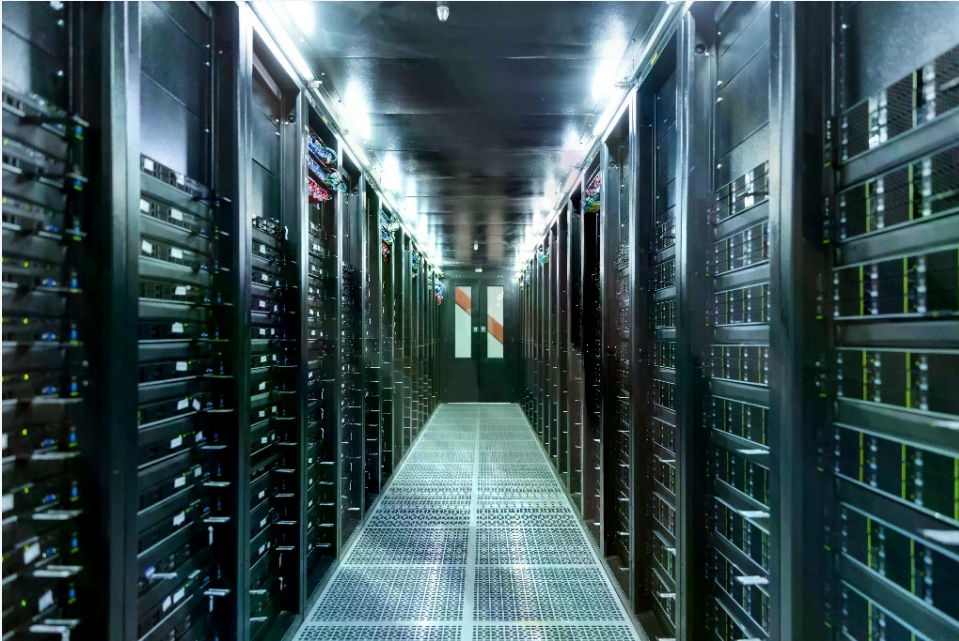
The foundation provides intelligent computing capabilities, high-speed networks, and distributed storage, supporting localized inference and fine-tuning of large models for one-stop services.

Integrating all campus business data to build high-quality corpora and business knowledge graphs, providing continuously evolving knowledge nourishment for agents.

Intelligent Computing · Knowledge Graph

MCP Protocol Empowerment: Key Technology to Break System Silos

MCP (Model Context Protocol) as a standard bridge connecting large models and campus underlying systems, has fundamentally changed the fragmented dilemma of traditional API integration. It encapsulates the capabilities of basic systems such as campus networks and libraries into standardized context tools, enabling agents to read data and execute operations safely and efficiently.

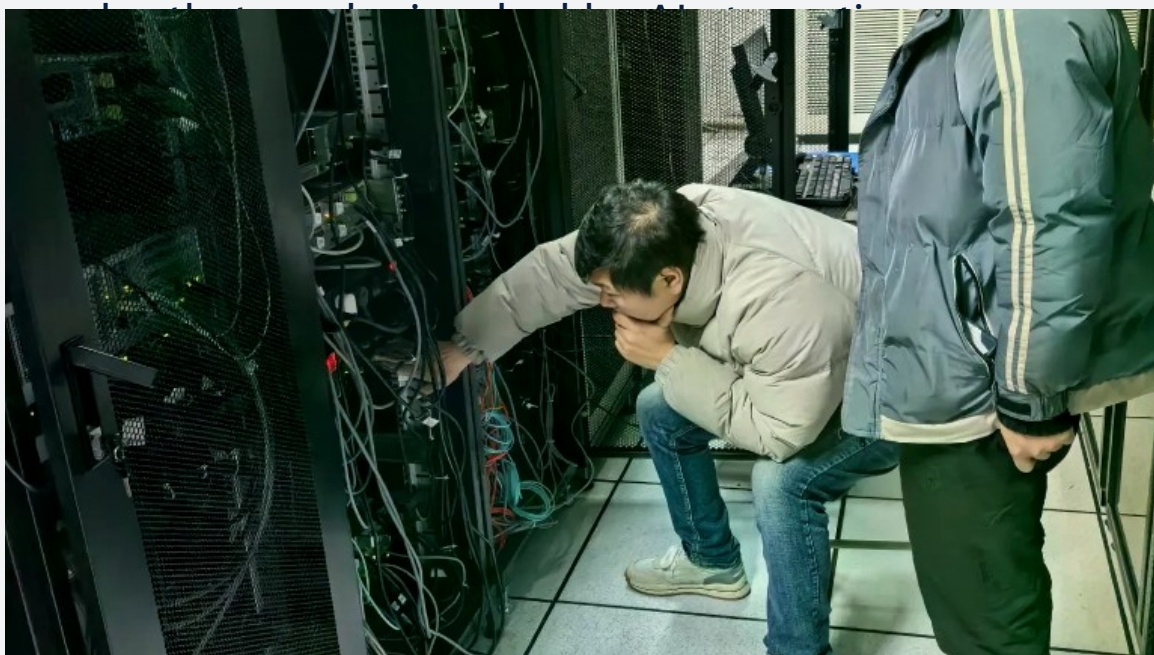


University Data Center · Campus IT Infrastructure Foundation

- 🔗 **Standardized Interface Specification:** The MCP protocol defines unified tool description and invocation specifications, enabling large models to understand and invoke system capabilities through standard interfaces without writing custom code for each campus system.
- 🔧 **Dynamic Interactive Operations:** Through the MCP toolkit, agents can not only obtain static data but also execute dynamic interactive operations such as "querying campus card balance" and "reserving library seats," achieving true on-demand service.
- 📄 **Command-line Direct Connection Mode:** The combination of command-line tools and the MCP protocol provides large models with the most direct and secure system interaction method, effectively avoiding the high latency and error rates associated with traditional graphical interface automation.

Basic System Integration: MCP Toolkits for Campus Network and Library

In high-frequency use cases such as campus networks and libraries, the platform has built fine-grained toolkits through the MCP protocol. This not only enables agents to accurately respond to network troubleshooting and literature retrieval needs, but also transforms originally isolated backend management systems into intelligent service



Campus Network Operations Site

Campus Network Operations MCP Tools

Provides tools for network status diagnosis, IP address query, and bandwidth limit removal, supporting agents in automatically troubleshooting network. Integrates with billing and authentication systems to achieve "one-sentence recharge" and "frictionless authentication," significantly reducing ticket pressure on IT service desks.



Smart Library Services

Library Service MCP Tools

Encapsulates capabilities such as collection search, e-book download, and loan period query, supporting agents in making personalized literature. Integrates space reservation and seat management systems, enabling intelligent allocation and conflict coordination of study room resources through natural language instructions.

CHAPTER 03

Security Control and Key Management Mechanism

Building Campus Data Defense in a Zero-Trust Environment

Access Control and Authentication System under Zero-Trust Architecture

With the rapid development of cloud computing and big data processing capabilities, the campus open platform must establish strict zero-trust security boundaries. Through multi-dimensional identity authentication and fine-grained permission control, agents always follow the "principle of least privilege" when invoking core campus systems, preventing data leakage and unauthorized operation risks.

Dual Access Control

RBAC + ABAC

Mechanism
Introduce dual access control based on roles and attributes, dynamically calculating agent operation permissions based on identity, department, and scenario. The system evaluates user identity attributes and resource sensitivity in real time, achieving fine-grained dynamic permission allocation.

Human-AI Collaborative Review

High Risk

Mechanism
High-risk operations such as financial reimbursement and grade modification forcibly trigger secondary manual authorization, ensuring absolute security at key business nodes. The review process is embedded in the agent execution chain, forming dual guarantees of automation and manual supervision.

Unified Gateway Authentication

MCP Gateway

All MCP tool invocations are authenticated through a unified gateway, comparing context and security policies in real time, automatically intercepting abnormal frequency and unauthorized access. The gateway layer integrates audit logs, completely recording agent operation trajectories to support post-incident traceability analysis.



University Cybersecurity Monitoring Center · Real-time Data Dashboard

API Key Full Lifecycle Management and Audit Tracking

Agents use the "digital passes" for agents to connect to campus systems. The platform has built a full lifecycle management closed loop covering key generation, distribution, rotation, revocation, and auditing, ensuring every system interaction is traceable and verifiable, providing financial-grade security for campus data assets.



Key Generation

Uses asymmetric encryption algorithms to generate unique key pairs for each agent instance, supporting setting key validity periods and invocation quotas by business scenario. PSA



Auto Rotation

Establishes an automated key rotation mechanism that seamlessly triggers key updates without service interruption when potential leakage risks are detected or lifecycle thresholds are reached. Auto



Blockchain Audit

Detailed records of each API invocation's timestamp, caller identity, request parameters, and response status, forming tamper-proof blockchain audit logs. Log



Security Dashboard

Provides a visual security dashboard for real-time monitoring of key usage anomalies and traffic spikes, supporting one-click circuit breaking of high-risk agent system access permissions. Live

Chapter 04

Agent Ecosystem and Future Evolution

Toward an Open, Collaborative and Continuous Growing Digital Workforce
Network

Building a University-level AI Center and SKILLS Plaza

The ultimate form of the campus open platform is a thriving agent ecosystem. By building a university-level AI center and SKILLS Plaza, the platform centrally opens up dispersed computing, model, and tool capabilities, encouraging students and faculty to independently create personalized agents based on the MCP protocol, achieving an ecological leap from "unilateral school provision" to "co-construction and sharing."

01 University AI Center Unified Management

Unified management of intelligent computing resources and base large models, providing low-threshold quotas for all students and faculty, avoiding duplicate "silo-style" AI application construction across departments.

02 SKILLS Plaza Open Aggregation

As the "app store" for campus agents, it aggregates various MCP tools and Agent templates subject to one-click subscription and flexible combination.

03 Capability Assessment and Review Assurance

Establishes agent capability assessment and listing review mechanisms to ensure each toolkit complies with campus data security regulations and business logic standards, guaranteeing healthy and orderly ecosystem development.



University Makerspace · Student-Faculty Teams Co-building Agent Ecosystem

Agent "Taming" Mechanism and Human-AI Collaborative Responsibility Chain

Agents are not perfect and cannot stand on their own. They require continuous "taming." Universities must establish optimization mechanisms based on real business feedback and clear responsibility boundaries, allowing agents to continuously evolve within a safe and controllable framework, ultimately achieving a new paradigm of

"Taming" and Evolution Mechanism

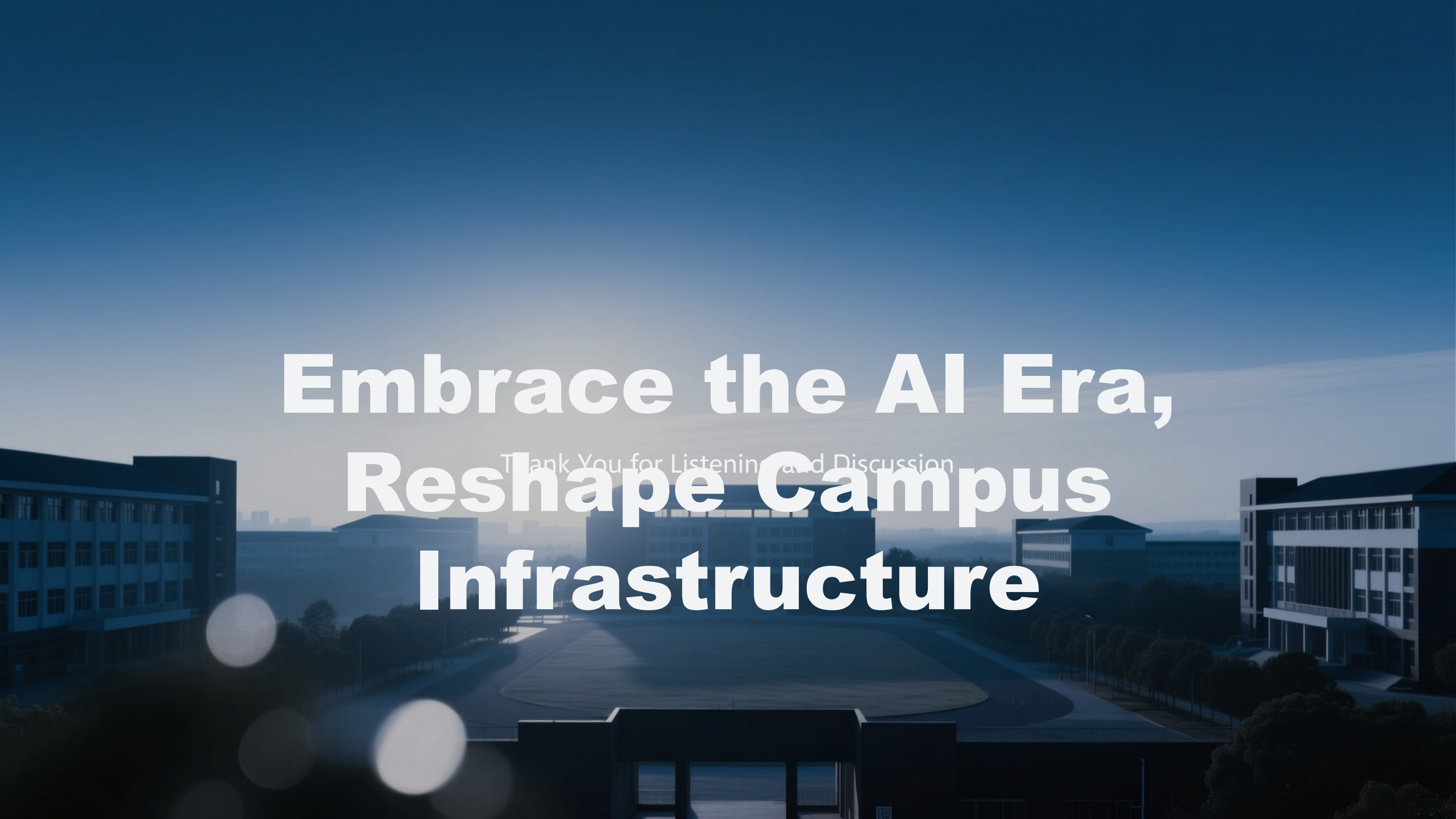
- 01 Establishes a task negative list, clearly defining that in high-risk scenarios (such as academic evaluation and disciplinary action), agents can only provide auxiliary suggestions, and automatic decision-making is strictly prohibited.
- 02 Introduces Reinforcement Learning from Human Feedback (RLHF), using likes or correction data on agent responses to continuously fine-tune localized business large models.

RLHF · NEGATIVE LIST

Responsibility Chain and Compliance Governance

- 01 Establishes a human-AI collaborative responsibility chain, ensuring manual review nodes always exist in key business workflows, with important results traceable to specific responsible persons and agent versions.
- 02 Coordinates development and security, establishes special prevention mechanisms for algorithmic discrimination, privacy leakage, and academic fraud, and regularly conducts agent ethics and security audits.

HUMAN-IN-THE-LOOP · AUDIT



Embrace the AI Era, Reshape Campus Infrastructure

Thank You for Listening and Discussion